

Characterization over time and growth model for the submarine cable network

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ABSTRACT

Around 99% of the global Internet traffic is estimated to pass through fiber optic cables laid down under the sea. These submarine cables together can be considered as a single network that has evolved over time, becoming a fundamental cornerstone of today's global Internet infrastructure. This work provides a detailed characterization of the evolution of this network, which grew from just ten to more than ten thousand nodes in nearly four decades. In addition to this large growth, peculiar invariant network properties are identified such as the absence of triangles, a large number of small components, and components in the form of simple paths. The peculiarities found in the SCN evolution are the motivation for a novel network growth model that can better represent this kind of network. The proposed model is simple having only two parameters and uses growth mechanisms designed to capture the evolution of the SCN, such as mixing preferential attachment with stretching paths. A direct comparison with real data and other models for network growth indicates that the proposed model is relatively accurate (under proper parameterization) and superior to alternative models.

KEYWORDS: submarine cable network; network characterization; network over time; network growth model.

1. INTRODUCTION

Submarine cables have played a crucial role in international communication systems since the late 19th century, when the first telegraph line was laid down across the English Channel, connecting London to the European telegraph network through Paris [1]. Currently, it is estimated that more than 99% of international and intercontinental telecommunication traffic rely on fiber optic cables installed under the sea (submarine cables) [2]. Internet traffic began to be carried over submarine cables in the late 1980s. Today, there are more than 600 submarine cables owned by hundreds of companies that provide Internet connectivity between cities, countries, and even continents around the world. Although this infrastructure represents a fundamental and critical cornerstone of the Internet and continues to grow every year, there are relatively few academic studies on its role in determining the cost, performance, and robustness of the global Internet [3].

The submarine cable network (SCN) is formed by considering all existing submarine cables since links of these cables can be installed at shared landing points (e.g. cities). TeleGeography (a

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telecommunications market research company) provides data on deployed submarine cables [4] as well as the date when each cable was ready for service, enabling the reconstruction of the SCN over the years, allowing for a detailed characterization of the evolution of this network.

Although recent work has analyzed different aspects of the SCN over time, their focus is rather different from the one taken in this work. For example, Xie *et al.* characterize the evolution of the SCN with respect to China and its relationship with investments and the economic development of its cities [3]. Bischof *et al.* focuses on characterizing metrics such as cable capacity and cable length over time as well as events responsible for cable failures [5]. In contrast, this work focuses on the evolution of the SCN from the perspective of metrics related to network structure, such as connected components, degree distribution, and diameter, to name a few.

The temporal characterization of different network metrics made in this work identified peculiar invariant properties in the SCN, such as the absence of triangles (and very few cycles), a large and growing number of small components in the form of simple paths (end nodes with degree 1 and all intermediate nodes with degree 2), and a persistent average degree of approximately 2. This characterization provides a general understanding of the evolution of the SCN and motivates the development of a network growth model that can adequately represent this network over time.

Most of the network growth models proposed in the literature generate connected networks with cycles and triangles. However, SCN is made up of many components and very few cycles due to its inherent nature of being a collection of submarine cables. Moreover, many of these components are simple paths. This motivates the development of a novel network growth model to accurately represent the SCN over time. The proposed growth model is based on three growth mechanisms: adding connected components, stretching components, and preferential attachment (*PA*). These mechanisms capture the essence of the evolution of submarine cables: adding a new cable (not connected to any other cable), adding a cable to an end point of an existing cable (stretching it), and attaching a cable to a popular landing point. The add-stretch-attach (*ASA*) model has just two parameters that determine the probabilities of adding a new component and stretching an existing cable or connecting it to a popular landing point via *PA*.

A direct comparison with real data and other models for network growth (adapted to generate multiple connected components) indicates that the proposed model is relatively accurate (under proper parameterization) and superior to alternative models. Extensive numerical results support these findings.

The remainder of this article is organized as follows. [Section 2](#) briefly describes recent work related to the SCN. A detailed characterization of the SCN constructed from the data over time is described in [Section 3](#). [Section 4](#) describes the *ASA* model in detail and a direct comparison between SCN and different networks growth models, including the one proposed in this work, is presented in [Section 5](#). Lastly, some final remarks are presented in [Section 6](#).

2. RELATED WORK

The characterization and evolution of the SCN date back at least 25 years ago with the seminal work of Beaufile [6], comparing the network created by this technology with the satellite network. Network characteristics such as link bandwidth and link usage were also analyzed over time, even before the cables were fiber optic. Malecki and Wei [7] also analyze the evolution of SCN 16 years ago. Their analysis shows the evolution of the SCN with respect to the number of cables, links, landing points, and bandwidth over time. In addition, they classify countries using these metrics in order to identify the countries that are at the center of the network. More recently (7 years ago) Bischof *et al.* [5] in addition to analyzing the evolution of cable length and its capacity, declared the need to characterize the SCN as a research agenda since this network plays a major role in modern communication infrastructure.

Several other works involving SCN have appeared in the literature, beyond its characterization over time. For example, some works deal with the path planning problem, which consists of adding new cables to the network taking into account costs and revenues combined with a series of other

factors related to survivability and robustness [8–13]. Another goal in path planning is to add new cables to improve network performance in order to maximize total throughput per fiber [14–16].

While obtaining information about the submarine cable network is relatively straightforward nowadays, using this data to support network decisions from the perspective of network operators is challenging [17–19]. In addition to investigating how to translate public announcements for submarine cables into network information (and the impact on the network), Livadariu *et al.* [18] propose a methodology to automatically extract IP-level information for new cables. Ramanathan *et al.* [19] introduce a framework to create a cross-layer map of submarine cables and IP links, i.e. the framework establishes a correspondence between IP links and submarine cables (when they traverse a submarine cable). Similarly, Carisimo *et al.* [17] present a methodology to identify long-haul IP links (links with high latency) that traverse submarine cables.

This work focuses on the analysis of network properties over time and, on the basis of these, presents a growth network model to represent the evolution of the SCN. To our knowledge, this is the first model proposal specifically created for the SCN.

3. SUBMARINE CABLE NETWORK

The SCN is a communication network made up of submarine cables that connect cities around the world through fiber optic cables installed on the seabed. Currently, the main purpose of this network is to carry Internet traffic between cities, especially those separated by the oceans. This physical network comprises devices located on the coast (dry plant) and below the sea (wet plant). In addition to fiber optic cables, the wet plant contains repeaters and branching units that are installed along the undersea cable. While repeaters are responsible for extending the signal's range, branching units are used to split the signal from a trunk cable into two other cables to carry data to different destinations. The copper conductor surrounding the fiber optic cables provides the necessary electrical power from the transmission terminal (dry plant) to these devices [1]. On the other hand, the dry plant is composed of several other digital devices which together are responsible for transmitting and receiving optical signals that correspond to Internet's traffic carried by submarine cables, here simply called as transmission terminal for simplification purpose.

The submarine cable map provided by TeleGeography [20], indicating the position of all current submarine cables (in different colors). In this map, each submarine cable system is shown as an independent network that connects two or more cities located on the coast of the same country or on different continents. The points on a cable shown on the map are called landing points. If a landing point is located on the coast, it represents the city's transmission terminal. In contrast, landing points located in the middle of oceans represent repeaters or branching devices.

Different submarine cable systems can install their transmission terminals at the same landing point. For example, there are ten submarine cables installed in Rio de Janeiro, assuming that all these cables share this landing point. Therefore, the SCN nodes considered in this work correspond to landing points where at least one submarine cable is installed, and each edge indicates the existence of a direct optical link between two landing points belonging to a certain submarine cable. For example, on the SAC cable, the landing points corresponding to the cities of Rio de Janeiro and Santos are connected by two repeaters. Since different cable systems can have links to the same pair of landing points, the network also allows multiple edges between two nodes. Thus, given a set of submarine cable systems, an SCN is constructed that represents them (their landing points and the direct connections between them). In the following, this network will be characterized over time.

3.1. Characterization of SCN over time

The SCN is characterized over time by analyzing different networks constructed independently with data provided by TeleGeography and extracted in January 2025. For each year from 1989 to 2028, an SCN is built considering cables ready for service or planned to be installed each year, giving rise to 40 different SCNs (one per year). The available data have a year granularity, and thus the choice of year timescale for analyzing the evolution of the SCN over time. Note that the set of cables active in

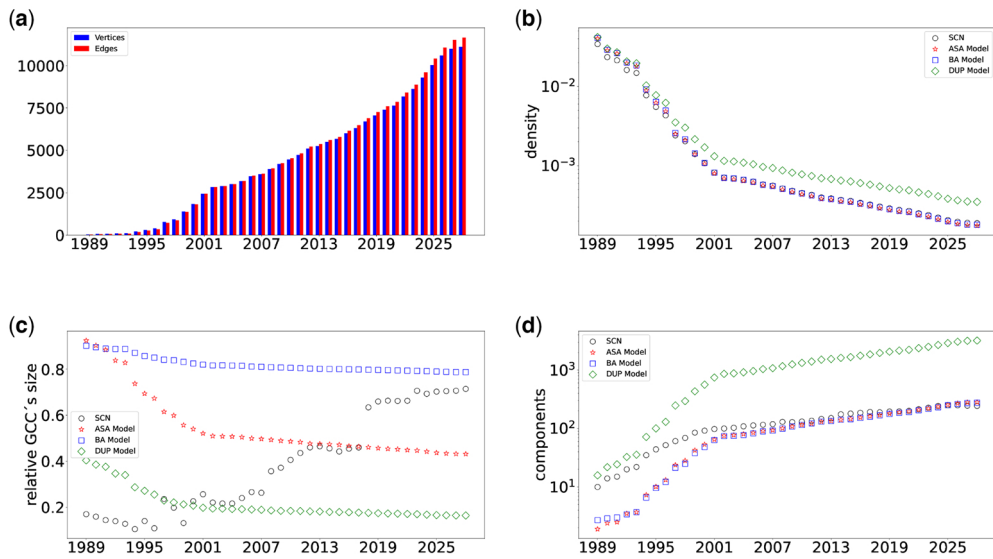


Figure 1. Vertices and edges of the SCN (a), edge density (b), GCC's relative size (c), and number of components (d) of the SCN and network growth models over time.

a given year is almost always different from the set of the previous year, as new cables are installed or expanded. Moreover, TeleGeography reports cable systems not only installed, but also cable systems in the process of deployment or planned for future deployment, which is why the data extend to 2028.

Figure 1a shows how the number of vertices and edges of the SCN has evolved over the years. As can be seen, the SCN has experienced substantial growth going from around 10 vertices to almost 10 000 vertices in 40 years. Notice that until the early 2000s, the number of edges was smaller than the number of vertices, as most cables connected only two landing points (one edge for two vertices). However, this trend reversed in the subsequent years, with the number of edges surpassing the number of vertices. Despite this increase, the density of the SCN (fraction of edges present) decreased significantly over the years, as shown in Fig. 1b, showing that the SCN becomes more sparse every year, as occurs in other real-world networks (Plots in this section have curves for data representing the SCN and growth model networks to be presented in Section 5.).

Despite being sparse, most real-world networks are almost completely connected since the majority of their nodes belong to the same component, known as the Giant Connected Component (GCC). The evolution of the relative GCC's size for the SCN is shown in Fig. 1c. Although it has its ups and downs, overall the GCC's size has been growing over the years and currently contains nearly 75% of the nodes. Therefore, the SCN has become more connected over the years, as most of the nodes are in the GCC, although the network is still very sparse and consists of many connected components, as shown in Fig. 1d. Indeed, this figure shows that the number of connected components in the SCN grows monotonically over time, indicating that many new cables do not share landing points with the previous cables that form the GCC. Interestingly, both the GCC's size and the number of connected components grow over the years. Today, about 25% of SCN's nodes are spread across more than 250 components (other than GCC).

Figure 2a shows the evolution of the average path length (APL) over the years. Note that the APL has an increasing trend over the years but with a decrease in some years, notably the last five. Its current value is around 60 (in 2025). This suggests that the SCN has expanded in size and structural complexity. Figure 2b shows how the largest distance between any pair of nodes in the GCC (the GCC's diameter) has evolved over the years. Again, there is an increasing trend in the GCC diameter

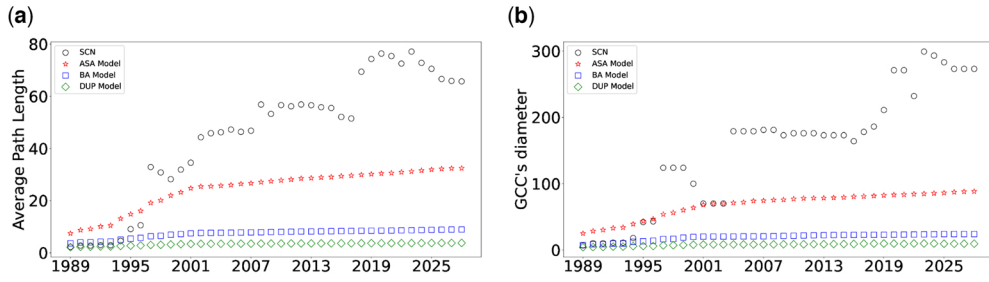


Figure 2. APL (a) and GCC's diameter (b) of the SCN and network growth models over time.

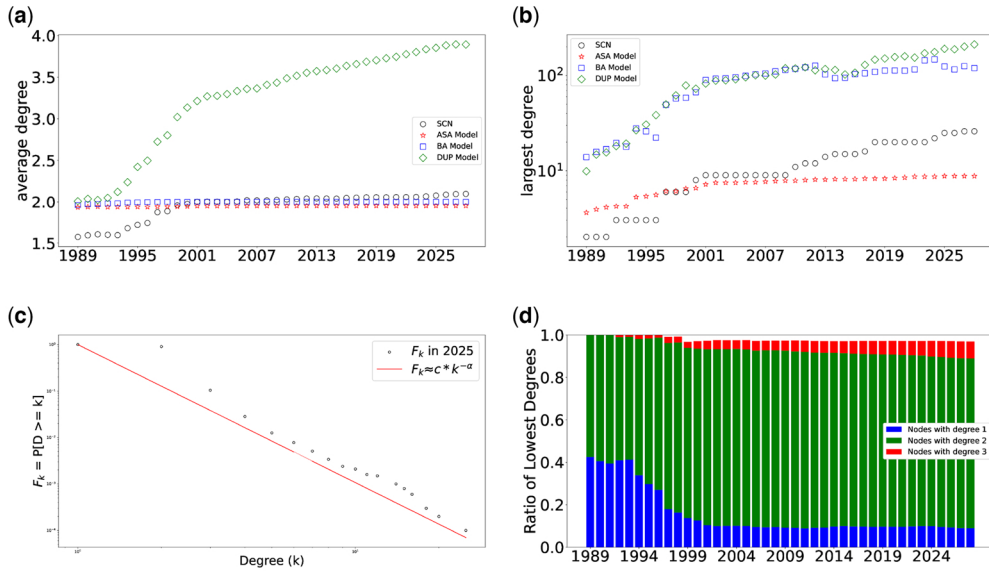


Figure 3. Average degree (a) and largest degree (b) of the SCN and growth models over time; CCDF for the degree of the SCN in 2025, shown in log-log scale (c), and smallest degrees (d) of the SCN over time.

over the years but with a decrease in some years. In fact, the evolution of the GCC's diameter is similar to the evolution of the APL (in Fig. 2a). Interestingly, the GCC's diameter did not change substantially from 2004 to 2015 (as well as the APL), while the number of nodes almost doubled in the same period, suggesting that the network contains *hubs*, i.e. nodes with large degrees (much larger than average) that maintains the network connected with small distances (and diameter).

The evolution of node's degrees in the SCN reveals some unique properties of the SCN. For instance, Fig. 3a shows the evolution of the average degree. This figure shows that it has remained practically constant for the past 30 years, with an approximate value of 2. Interestingly, the largest degree of the SCN evolves very differently, as shown in Fig. 3b. Since 2000, the largest degree was already four times larger than the average degree. Furthermore, the largest degree continued to grow, albeit slowly over time, reaching the value of 26 in 2028, being more than ten times larger than the average degree, suggesting that the degree of the largest *hub* is also increasing over time.

The degree distribution may indicate the existence of *hubs* if it is a heavy-tailed distribution, showing the existence of nodes with degrees much larger than average. Figure 3c shows the complementary cumulative distribution function (CCDF) of SCN's degree distribution for 2025. It is possible to see that it is a heavy-tailed distribution due to the linear decay of the probability. More rigorously, we estimate the exponent around 3.0 for a power-law distribution that best fits the

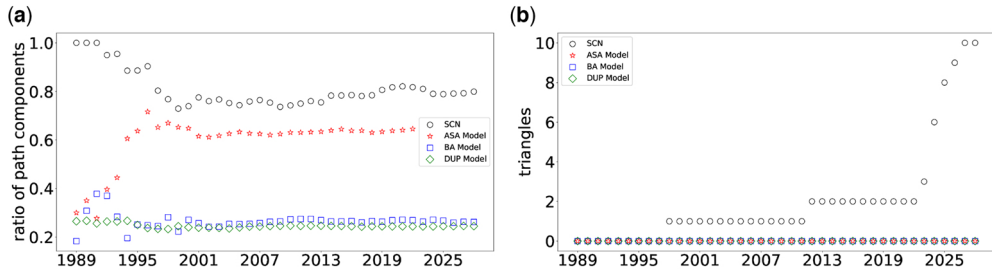


Figure 4. Fraction of components that are simple paths (a) and number of triangles (b) of the SCN and network growth model over time.

observed data, which was obtained by linear regression using the ordinary least squares method. This suggests that the SCN is scale-free, despite having a limited range of degrees. Figure 3c also shows the theoretical linear function whose slope is the estimated exponent (3.0), indicating its agreement with the empirical SCN's degree distribution. Clearly, today the SCN is a network with *hubs*.

Figure 3d shows the fraction of nodes with the smallest degrees (degrees 1, 2, and 3) in the SCN over the years. Interestingly, the fraction of nodes with degree 1 has remained constant for the past 28 years, around 9%. During almost the entire period, the fraction of nodes with degree 2 is around 80%. In contrast, the fraction of nodes with degree 3 has increased slowly in the last 28 years, reaching around 8% in the last year. Note that only a small fraction of nodes (3%) have a degree larger than 3 in the current SCN, suggesting that there are very few *hubs* in this network.

Given that a large fraction of nodes have degree 2, a constant fraction have degree 1 and the fact that the network has many small connected components, it seems intuitive that some of these components are simple paths: two nodes with degree 1 (end nodes) and all other nodes with degree 2 (intermediary nodes). Figure 4a shows the fraction of components in the SCN that is a simple path over the years. As can be seen in this figure, the vast majority of components in the SCN are simple paths, in accordance with their inherent nature of representing submarine cables.

The clustering coefficient is one of the important characteristics of real-world networks, since they are directly related to network transitivity and robustness. This metric indicates the tendency of neighbors of a given node to be linked to each other, which can provide important clues about the presence of communities. To calculate the clustering coefficient, it is necessary to measure the number of triangles (cycles of length three). Unlike other real-world networks, the clustering coefficient of the SCN is insignificant. This can be seen in Fig. 4b, which shows that until 1998 the SCN had no triangles and from 1998 to 2011 it had only one triangle. Although in the current year there are 10 triangles, they can be considered negligible given the size of the network and the number of paths of size two.

4. THE ADD-STRETCH-ATTACH MODEL

The previous section showed that the SCN is composed of many connected components from which a large fraction (around 80%) are simple paths, while it also has a GCC containing approximately 80% nodes. The SCN has a heavy-tailed degree distribution with a large fraction of nodes (around 80%) with degree 2 and consequently the average degree approximately equal to 2. Moreover, the network is extremely sparse and the number of triangles (cycles of length three) is negligible. These findings motivate a novel network growth model based on the following three mechanisms: (i) add nodes to form new connected components; (ii) stretch paths in the network by connecting new nodes to nodes with degree 1 already in the network; (iii) attach new nodes to popular nodes by PA, i.e. nodes already in the network are chosen with probability proportional to

Algorithm 1 Generating a random network with the ASA model**Require:** n, p, q

```

1:  $t \leftarrow 1$ 
2:  $G_{t-1} \leftarrow \{\}$ 
3: while  $t \leq n/2$  do
4:    $g \leftarrow \text{new\_2clique}(2t-1, 2t)$ 
5:   if  $\text{uniform}(0, 1) \geq p$  then
6:     if  $\text{uniform}(0, 1) \leq q$  then
7:        $w \leftarrow G_{t-1}.\text{get\_node}(\text{deg} = \{1\}, \text{pa} = \text{False})$ 
8:     else
9:        $w \leftarrow G_{t-1}.\text{get\_node}(\text{deg} = \{\text{all}\}, \text{pa} = \text{True})$ 
10:    end if
11:     $x \leftarrow g.\text{get\_node}(\text{deg} = \{1\}, \text{pa} = \text{False})$ 
12:     $G_t \leftarrow G_{t-1}.\text{add\_link}(x, w)$ 
13:  else
14:     $G_t \leftarrow G_{t-1}.\text{add\_graph}(g)$ 
15:  end if
16:   $t \leftarrow t + 1$ 
17: end while
18: Return:  $G_t$ 

```

its current degree. Thus, the goal of the proposed Add-Stretch-Attach (ASA) model is to capture the essence of the SCN's growth and adequately represent its evolution over time.

Consider discrete time steps $t = 1, 2, \dots$ and let G_t denote the random graph at time t . At each time step t , a 2-clique (the simplest possible clique in a graph) (two vertices connected by an edge) is added to the existing graph G_{t-1} , giving rise to G_t . Vertices are numbered sequentially according to their arrival times, such that vertices $2t-1$ and $2t$ form the 2-clique added to the graph at time t . With probability P , the 2-clique created at time t is added to G_{t-1} as a new connected component (without edges to any existing vertex). But with probability $1 - P$, the newly created 2-clique will be connected to a vertex in G_{t-1} . The edge responsible for this connection will be created between the 2-clique and an anchor vertex chosen according to two distinct mechanisms. With probability q , it will be chosen uniformly at random from vertices of degree 1 in G_{t-1} (to stretch a path), and with probability $1 - q$ it will be chosen from vertices of G_{t-1} according to PA. Then, an edge is added between the anchor vertex and a random vertex from the 2-clique. For completeness, let $P_i(t)$ and $d_i(t)$ denote the probability that vertex i is selected at time t and the degree of vertex i at time t . $P_i(t) = I(d_i(t-1) = 1) / \sum_{j=1}^{2(t-1)} I(d_j(t-1) = 1)$ given the stretching mechanism, where $I(\cdot)$ is the indicator function, and $P_i(t) = d_i(t-1) / \sum_{j=1}^{2(t-1)} d_j(t-1)$ given the PA mechanism.

The [algorithm 1](#) describes the generation of a random network (G_t) according to the ASA model. The parameters of the algorithm are n , the number of vertices in the network to be generated (assuming that it is even) and the probabilities P and q as previously discussed. At each iteration, the 2-clique is created with vertices $2t-1$ and $2t$ linked by one edge (line 4) and added to G_{t-1} as a new component (line 14) with probability P . This new component can be connected to the existing network with probability $1 - P$ (line 5). In this case, the choice for the anchor node (w) in G_{t-1} can follow the stretching mechanism (with probability q , line 7) or the PA mechanism (with probability $1 - q$, line 9). Finally, a random vertex (x) in the g is selected and connected to the chosen anchor node (lines 11 and 12).

The function `get_node` in the [Algorithm 1](#) selects a node at random according to its two parameters. The first parameter *deg* determines the degrees of the nodes to be considered in the random

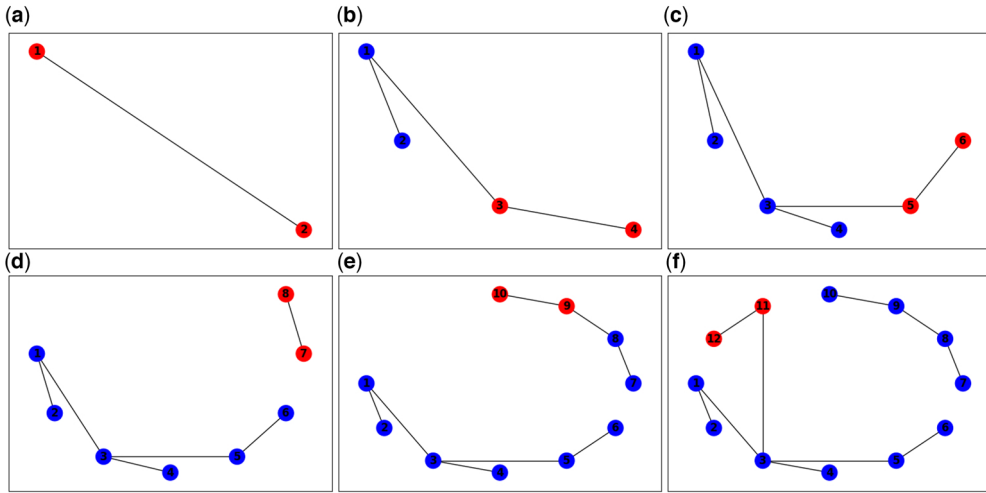


Figure 5. Example of network generated by the ASA model.

selection, while the second parameter pa determines if nodes should be chosen uniformly at random ($pa=False$) or proportional to their degree ($pa=True$).

Figure 5 illustrates the growth of a network using the ASA model. At each step, cliques of size 2 are added to the network. In the beginning (Fig. 5a), the clique 1-2 is added as a new component because there are no nodes in the network. Next, clique 3-4 is added to the network, and by chance, with probability q , was connected to an anchor node (node 1), as shown in Fig. 5b. Then, clique 5-6 is added to the network and connects to node 3, which has degree 2 and a larger chance under PA ($1-q$), Fig. 5c. Figure 5d shows that the clique 7-8 is created as a new connected component, which occurs with probability P . Figure 5e shows that clique 9-10 connected to node 8 stretching the path of this new component. Finally, to illustrate the appearance of *hubs*, Fig. 5f shows that clique 11-12 connected according to PA to node 3. Note that the degree of node 3 became twice the average, indicating that it is a *hub*.

Note that the ASA model adds two nodes and one or two edges at each time step. Thus, G_t has exactly $2t$ nodes and $t(2-P)$ edges in expectation, and therefore an average degree of $2t(2-P)/2t = 2-P$, which is independent of t . Another property is that G_t does not have cycles, as at most one edge is added between the new clique and the anchor node of the existing graph. Finally, each component grows independently of the other components, as they are never interconnected.

4.1. Characterization of ASA model

In addition to the number of nodes (n), the ASA model has only parameters P and q . As discussed, P controls the rate at which new connected components are created in the network, and q controls the ratio between stretching and PA mechanisms. Interestingly, when $P = 0$ and $q = 0$ the ASA model is almost equivalent to the Barabási-Albert model [21], and thus bound to generate networks that have a power-law degree distribution (The difference is that ASA model adds 2-cliques at each time step whereas the BA model adds a single node.). Moreover, if $P = 0$ and $q = 1$ the ASA generates a single path network where all nodes have degree two except the two leaf nodes that have degree one. Also note that if $P = 1$ then ASA generates a network with $n/2$ connected components each corresponding to a 2-clique, independently of q . Thus, the values of P and q play a fundamental role in the structure of the random graph G_t .

In order to understand the role of P and q in the ASA model, heatmaps are plotted in Fig. 6 showing the average value of different network properties for different values of P and q . Each

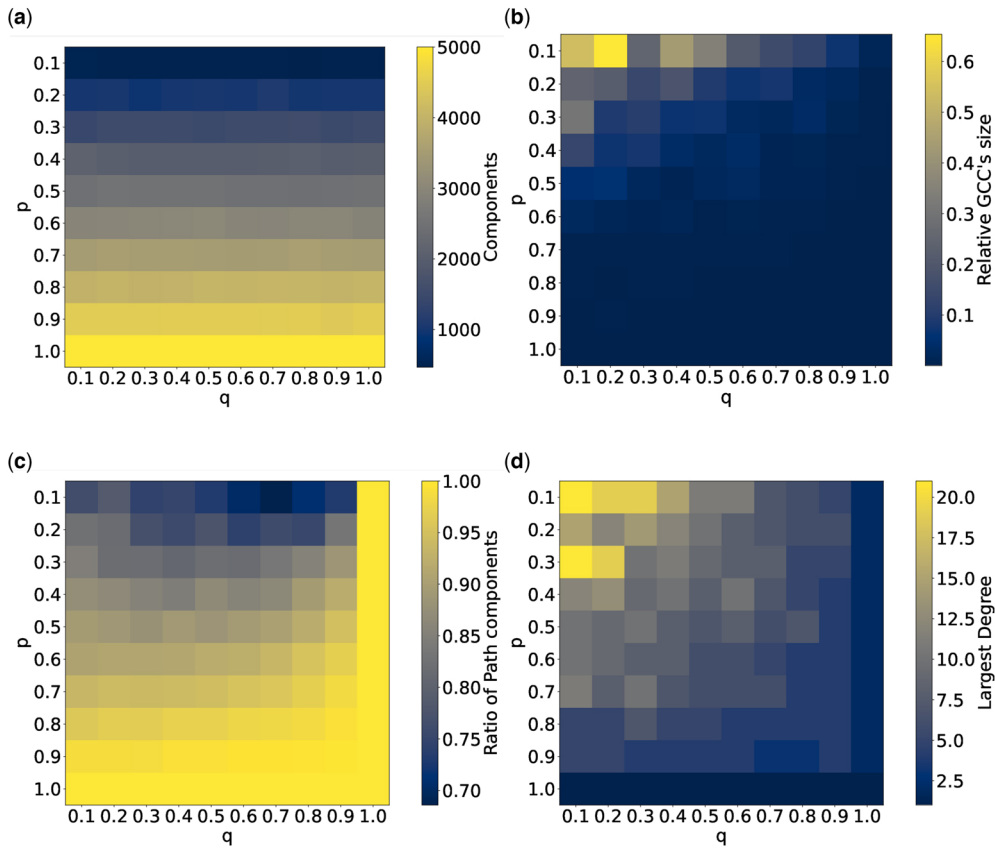


Figure 6. Heatmaps in function of P and q indicating the average number of components (a), average size of the GCC (b), average ratio of connected components that are simple paths (c) and average largest degree (d) (average over 100 independent samples of the ASA model).

heatmap was generated considering data from 100 independent networks, each with 10 thousand nodes generated by ASA. The values of P and q varied from 0.1 to 1.0.

The heatmap of the connected components in Fig. 6a shows that as P increases, the number of components increases. Furthermore, the value of q does not influence this metric, as expected. In contrast, Fig. 6b shows how the relative size of GCC is influenced by the values of P and q . Note that the largest GCC occurs when P is equal to 0.1. When $P \geq .5$, the GCC has less than 15% of nodes, but decreases monotonically with P . In this regime, q has little influence on the size of the GCC.

The parameter q plays a role in the ratio of simple paths and the largest degree. Figure 6c shows that the larger the value of q , the larger is the ratio of components that are simple paths. This is because the PA growth mechanism is used with probability $1 - q$ and this mechanism creates components that have nodes with degrees greater than two, thus breaking simple paths. Note that P also impacts this metric, and when P increases, more components appear in the network, and consequently a larger fraction of these components are simple paths (see Fig. 6c).

Another network property significantly influenced by q is the largest degree of the network, as shown in Fig. 6d. As q decreases, the largest degree increases, and this is because of the PA growth mechanism that is known to generate nodes with large degrees. However, P also influences the largest degree, as lower values of P generate fewer components and allow for arriving 2-clique nodes to connect to existing nodes. By connecting to existing nodes, the value of the largest degree can

increase. Thus, for a fixed q , decreasing P will increase the value of the largest degree, as shown in Fig. 6d.

Some network properties are not influenced by the values of P and q . For example, the number of triangles generated by the ASA model is always zero, independent of P and q . In fact, ASA never generates a cycle in the network because it connects a 2-clique to some existing node or creates a new component. Another example is the edge density of the network that does not depend on q . In fact, the average number of edges generated by the ASA model with n nodes is $n/2 * (P + 2(1 - P)) = n(1 - P/2)$. Thus, when $P = 1$ or $P = 0$ the number of edges is $n/2$ or $n - 1$, respectively. Thus, the average edge density of ASA is $\frac{2-P}{n-1}$. Note that the average density depends on P but only linearly and decreases to zero with n (since we divide by n).

5. COMPARISON OF GROWTH MODELS

The proposed ASA model is evaluated and compared with two other models to capture the growth of the SCN. The Barabási-Albert (BA) [21] and Partial Duplication (DUP) [22] models follow PA and duplication attachment (DA) as their growth mechanisms, respectively. While PA gives preference to creating connections with more popular vertices within a network, in DA, connections of new vertices are made by creating connections to neighbors of selected vertices. These models were chosen for comparison because they are well-established random network growth models in the literature that generate networks with a degree distribution that are heavy-tailed (in fact, follows a power-law distribution), as is the case with the SCN (which is certainly heavy-tailed). However, these models were adapted to allow the generation of multiple connected components, as is the case with SCN (details below).

In the experiments, all synthetic networks created with growth models had the same number of nodes as the SCN has each year, allowing to compare the evolution of such networks with the SCN's evolution in the year timescale. For standardization purposes, the initial network for all models is a 2-clique, ensuring that all synthetic networks start identically. Additionally, for each model, 10 independent networks were generated, and the average statistics for each year are reported.

5.1. Barabási-Albert model

Barabási and Albert [21] proposed a network growth model, known as the BA model, which incorporates the PA (preferential attachment) growth mechanism, giving preference to creating connections with more popular nodes within a network. In this model, at each time step, a new vertex is added to the network with m incident edges (m is a parameter of the model). The m neighbors nodes of the new node are randomly chosen from among the nodes in the network according to a probability that is proportional to their degree. Thus, a node with a degree k times larger than another will have k times more chance to be chosen as the neighbor. This mechanism will increase the degree of nodes that have larger degrees.

In order to allow the classic BA model to generate multiple connected components, we consider a variation of the model where each new node can be isolated with a self-edge (loop) with probability P . With probability $1 - P$, the new node is added according to the original model. Thus, the expected number of connected components generated by this variation when n nodes are added is $E[C] = np$. Thus, if c is the number of connected components in the SCN, then P can be determined as $P = n/c$. Taking into account the SCN in 2025, we have $c = 251$ and $n = 10029$ and therefore $P = 0.025$.

The number of nodes added by the BA model each year is given by the number of vertices in the SCN in the corresponding year minus the number of vertices in the previous year's SCN. The number of edges added with a new vertex (parameter m in the BA model) was set to $m = 1$. This is because the average degree of the SCN in 2025 is 2.1. As the expected average degree of the BA model is $2m$, m was assigned to 1 because it is the closest integer to reach 2.1. Note that with $m = 1$ there are no cycles (and thus no triangles) in the network and the connected components

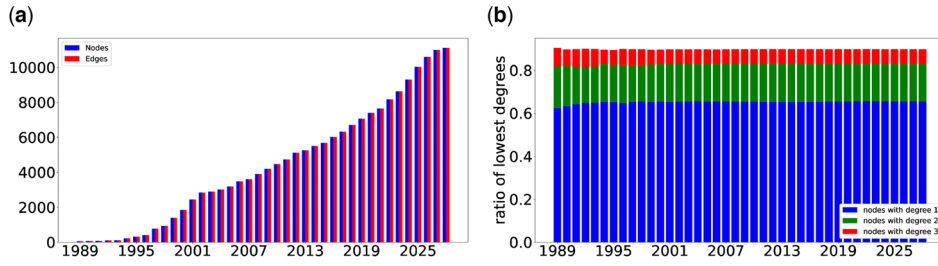


Figure 7. Nodes and edges (a) and smallest degrees (b) of the BA model over time.

are not merged by an incoming vertex. In what follows, this variation of the BA model is analyzed and compared with the SCN.

Figure 7a shows the evolution of the number of vertices and edges over time. Since $m = 1$, the number of edges is equal to the number of nodes for every year. Although showing a similar behavior, the number of edges in the SCN becomes slightly larger than the number of vertices over the years (see Fig. 1a). However, their density decreases with time at a similar rate, as shown in Fig. 1b.

While this variation of the BA model generates multiple components, it continues to generate a GCC. But, its relative size slowly decreases over time until it reaches around 80% vertices in recent years, while the relative size of the SCN GCC increases over the years and has approximately 70% nodes in recent years, as shown in Fig. 1c.

The number of connected components in this BA model grows linearly with the number of vertices, because P is constant and $m = 1$. This is similar to the growth of connected components in the SCN shown in Fig. 1d. Since P was chosen to match the number of components in the SCN in 2025, the two curves have been very similar over the last decade.

Figure 2b shows that the GCC's diameter in the BA model grows very slowly compared to the GCC's diameter of the SCN. For example, in the last 10 years, the GCC's diameter in BA model networks is around 25 while the GCC's diameter of the SCN is about 10 times larger.

The growth of the largest degree of BA model networks (and others) over time is shown in Fig. 3b, which reveal the emergence of *hubs*. However, the largest degree in BA model networks grows much faster than in the SCN. Thus, *hubs* generated by the BA model are much larger than in the SCN. In particular, in 2025, the largest degree in BA is about five times larger than that of the SCN.

Unlike the SCN, Fig. 7b shows that the majority of vertices in the BA model networks have degree 1 (around 60%), while in the SCN, most vertices have degree 2 (around 80%). Considering vertices with degree 3, both have an equivalent proportion, around 8%. Finally, few vertices in BA model networks have degree 2 (around 20%), which is very different from SCN. Thus, the BA model does not capture neither the smallest degrees (their distribution) nor the largest degrees (their values) of the SCN.

While this customized version of the BA model generates multiple connected components, the fraction of them that correspond to simple paths has been around 30% for the past 25 years, as shown in Fig. 4a. This differs significantly from SCN, where around 80% of the connected components are paths.

5.2. Partial duplication model

The partial duplication model is a growth model proposed by Chung *et al.* [22] (DUP model). In their proposal, a network is constructed by adding one node at time. When a new node (v) is created, an anchor node u is chosen uniformly at random from the existing network. Next, v is connected to each neighbor of u with probability P , independently. Note that v can also enter the network as an isolated node (with degree 0) if it does not connect to any neighbor of u , thus creating a new

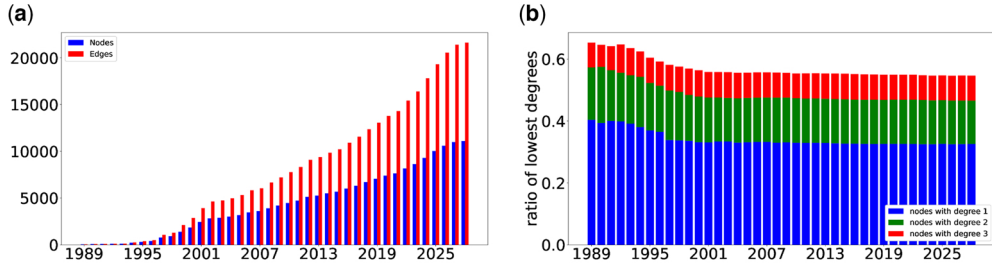


Figure 8. Nodes and edges (a) and smallest degrees (b) of the DUP model over time.

connected component, which occurs with probability $(1-P)^{d_u}$, where d_u is the degree of the anchor node (node u). But such nodes will always be isolated since they have no neighbors.

Different from the original DUP, we consider a variation in which if the anchor node u has no neighbors, the new node v could also be connected to u with probability P too. With probability $1 - P$, the new node v also enters the network as an isolated node with degree 0. Note that this variation allows for the growth of isolated nodes as new connected components.

In order to create networks to model the evolution of SCN with the DUP model, it is required to define P using SCN's data. Considering two networks with one year difference (G_{t-1} and G_t), we can extract from each network the number of nodes ($n(t)$) and edges ($m(t)$) for each year. Since the SCN at time t is an evolution of the network at time $t - 1$, the number of nodes and edges added by the DUP model to the network at time t is the difference in the SCN data between t and $t - 1$, respectively.

Note that the expected number of edges created by a new node in the DUP model that enters at time t is $P\bar{d}_{t-1}$ where $\bar{d}_{t-1}$ is the average degree of the network at time $t - 1$. We assume that all new nodes entering a given year experience the same average degree. Thus, $P\bar{d}_{t-1}(n(t) - n(t - 1))$ is the total of edges added by the number of nodes entered in year t . Moreover, note that $\bar{d}_{t-1} = 2m(t - 1)/n(t - 1)$, by definition. This should be equal to the number of edges added to the network at time t , in particular $m(t) - m(t - 1)$. Thus, the parameter P for a given year t can be defined as follows:

$$P = \frac{n(t - 1)(m(t) - m(t - 1))}{2m(t - 1)(n(t) - n(t - 1))} \quad (5.1)$$

Using SCN's data in 2024 ($n = 9295$ and $m = 9608$) and 2025 ($n = 10029$ and $m = 10419$), the value for P is approximately equal to $P = 0.53$ according to Equation 5.1. Since the model has a fixed value for P , that does not change over time, this value was used throughout the years.

Figure 8a shows the number of vertices and edges of DUP model networks over time, indicating that it has many more edges than vertices, which is different from the SCN where the number of edges is only slightly greater than the number of vertices. Consequently, the density of DUP model networks falls by a less pronounced rate than in the SCN over the years (Fig. 1b). Despite this difference, DUP model networks are also sparse.

When comparing the evolution of connected components of SCN and DUP model networks in Fig. 1d, it is possible to see that DUP model generates networks with a huge number of connected components compared to SCN. For example, this figure shows that in early 2000s DUP model networks have hundreds of connected components, while the SCN had less than a hundred components in the same period.

Different from the SCN, nodes in DUP model networks are widely distributed across its connected components. According to Fig. 1c the GCC's size of DUP model networks begins with less than 20% of nodes and decreases over the years, reaching approximately 10% of nodes in recent years. This is very different from SCN where this proportion increases over the years and currently

80% of the nodes belong to the GCC. Regarding GCC's diameter, DUP model networks have low diameter, especially because the GCC has fewer nodes. Over time, the GCC's diameter in DUP model networks remained below 10 while in the SCN, the GCC's diameter has increased over the years and reaching approximately 300 in recent years (Fig. 2b).

As with prior models, DUP model does not generate triangles. This is very similar to what is found in the SCN which, despite having few triangles over the years, can be said to be negligible (Fig. 4b).

The average degree of DUP model networks are smaller than the average degree found in the SCN in the early years, as can be seen in Fig. 3a. However, from 1992 onwards, this difference was reversed and the average degree of DUP model networks became greater than that of the SCN and remained so over the years, increasing this difference each year until reaching approximately twice the average degree of the SCN in 2025.

The Fig. 3b shows that the largest degree of DUP model networks partially follow the growth trend of BA model networks, increasing over time and substantially larger than the largest degree in SCN. So, DUP model networks have *hubs* with much larger degrees compared to *hubs* in the SCN.

Different from SCN, DUP model networks have around 15% of nodes with degree 2 while the SCN has approximately 80% (Fig. 8b). On the other hand, almost 35% of nodes have degree 1 in DUP model networks while less than 10% have such degree in the SCN. In addition, as shown in Fig. 8b, nearly 40% of nodes of DUP model networks have degree different from 1, 2 and 3. This does not mean that such nodes are *hubs*, having degrees greater than 3. but because this model allows the generation of nodes with degree zero.

Although it has many more connected components, Fig. 4a shows that the fraction of them that correspond to paths in DUP model networks evolves similarly to the BA model networks, and thus, about four times smaller than the fraction of connected components observed in the SCN.

5.3. ASA model

In order to compare ASA model networks with the SCN, it is necessary to determine ASA model parameters, namely P and q . Their values will be determined using the SCN's data, as follows.

Let C be the (random) number of connected components generated by the ASA model for a network with n nodes. Note that the expected number of connected components is given by $E[C] = np/2$, because for every 2-clique added to the network, a new connected component is created with probability P . Thus, if c is the number of connected components in the SCN, P can be determined as follows:

$$P = \frac{2c}{n} \quad (5.2)$$

Consider the probability that the ASA model generates nodes with degree 2 when the 2-clique is added to the network. With probability P no node with degree 2 is created. On the other hand, with probability $1-P$, the newly created clique will be connected to some existing node in the network. Then, with probability q , one node from the 2-clique will be connected to a existing node with degree 1 (stretching a component). In this case, two nodes will have degree 2 after such connection: one from the clique and one from the existing graph. On the other hand, with probability $1-q$, at least one node will have degree 2 (one of the clique nodes). Note that it is unlikely that the chosen anchor node has degree 1, since PA will give low probability to such nodes. Hence we consider that only one node will have degree 2 in this case. Therefore, assuming that $n/2$ cliques are added to the network, the average number of nodes with degree 2 can be approximated by the following equation:

$$D_2 = \frac{n}{2} (2(1-P)q + (1-P)(1-q)) \quad (5.3)$$

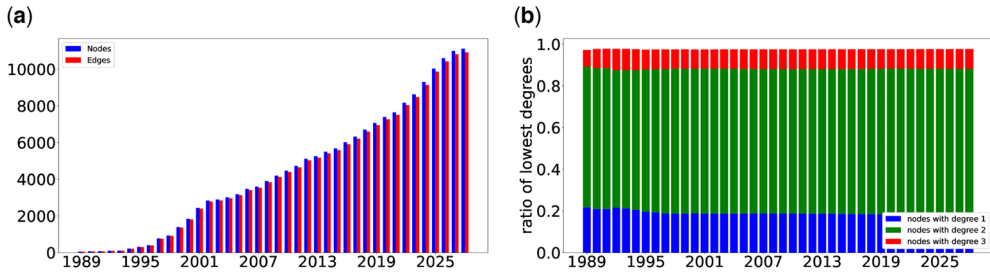


Figure 9. Nodes and edges (a) and smallest degrees (b) of the ASA model over time.

The probability q can be obtained analytically manipulating [equation 5.3](#), and is given by

$$q = \frac{2D_2 + n(P - 1)}{n(1 - P)} \quad (5.4)$$

Thus, q is determined using n , P and d_2 , the number of nodes with degree 2 in the SCN. Using real network data in 2025, we have: $c = 251$, $d_2 = 8027$ and $n = 10029$. Therefore, applying equations 2 and 4, we obtain $P = 0.05$ and $q = 0.69$, respectively, which are the parameters used in the ASA model to create networks for all years.

As well as the other models used in this comparison, ASA model generates extremely sparse networks like the SCN, as can be seen in [Fig. 1b](#), which shows the density of the network over time. This can also be seen in [Fig. 9a](#), which shows that the ASA model networks have fewer edges than the nodes over time.

Considering the evolution of the GCC, [Fig. 1c](#) shows that the GCC's size of ASA model networks gradually decreases over the years and, in the last year, has approximately 45% of nodes. In relation to its diameter, [Fig. 2b](#) shows that the GCC's size of ASA model networks has a relatively smaller diameter compared to the SCN's GCC for all years. Although the ASA model did not capture the GCC's properties found in the SCN, such a model compared to BA and DUP models better represents the SCN. For instance, the GCC's sizes of SCN and ASA model networks are similar between the years 2000 and 2015 while the other models did not produce values close to SCN for similar periods. Moreover, ASA model produces networks whose GCC's diameter is bigger than the GCC's diameter from networks generated by BA and DUP models for all years.

An advantage of the ASA model is that it produces networks with multiple connected components in the same proportion as found in the SCN, which can be confirmed in [Fig. 1d](#). Only in the early years the number of connected components between SCN and ASA model networks was different. But since early 2000s such values are very close. In addition to ASA model, only the BA model also captured this property of the SCN.

As in the other models analyzed in this work, ASA model generates networks with no triangles throughout all the years, which is similar to what happens in the SCN almost every year ([Fig. 4b](#)).

ASA model produces networks with an average degree similar to SCN, although slightly lower in almost the entire period ([Fig. 3a](#)). In this sense, DUP model was the only one that did not capture this property well. Despite the largest degree of ASA model networks be relatively smaller than those found in the SCN ([Fig. 3b](#)) for almost every year, such degrees can be considered closer to what can be found in the SCN. Comparing with the largest degree of networks generated with the BA and DUP models, it is possible to see that such models generate much larger degrees than those found in the SCN. Therefore, in this criterion, we consider that the ASA model is more suitable than the others. [Figure 9b](#) shows the proportion of lowest degrees found on ASA model networks. Approximately 70% of nodes have degree 2, a little less than what is found on SCN (80%). It also generate nodes with degree 3 in the similar proportion than in the SCN (around 8%). However, when considering nodes with degree 1, the ASA model has around 18% while the SCN has around

8%. Compared to BA and DUP models, ASA model is more suitable for modeling the SCN in relation to the proportion of lowest degrees, especially considering nodes with degree 2.

Finally, Fig. 4a shows that the majority of connected components in ASA model networks are simple paths, converging to about 70% over the last 25 years, despite the increase in the number of connected components. This is a significant advantage between ASA and the other models because the fraction of components in ASA model networks that are paths is much closer those found on SCN where simple paths is around 80%.

6. CONCLUSION

The submarine cable network (SCN) is a network built up with collection of fiber optic cables installed at seabed in order to interconnect cities, countries and continents, playing a crucial role in today's Internet communication infrastructure. Using publicly available data concerning submarine cables that have been deployed over the last 40 years, this work first presents the characterization of the SCN's structure over time. Results show that the network is formed of many connected components while also having a giant connected component whose size and diameter increases over the years. Its average degree is around 2 but the largest degrees grows over time, and the network exhibits a heavy-tailed degree distribution indicating the presence of *hubs*. Interestingly, a large fraction of the connected components are simple paths and there is essentially no triangles in the GCC, contrary to what might have been expected as a real-world network.

In light of the characterization of the SCN, a second contribution of this work is a novel network growth model, named Add-Stretch-Attach (ASA) model. The model embodies three growth mechanisms: adding new connected components, stretching connected components, and performing PA. These mechanisms reflect the particularities of SCN in a parsimonious model with just two parameters. The ASA model is compared to simple variations of the BA and DUP models (that can generate multiple connected components). Results indicate that ASA is more suited than the others in capturing the growth of the SCN with respect to different network characteristics, such as the fraction of connected components that are paths, and the proportion of lowest degrees. However, the ASA model (and others) fails to capture the increase in the relative size of the GCC, as well as its large diameter. This possibly occurs because some connected components are merged to the GCC over time, a feature not embodied in the ASA model and topic for future investigation.

Finally, while the ASA model is based only on structural characteristics, economic factors of cities and their bandwidth demands also play a role in shaping the evolution of the network (see [3], for example). However, incorporating such factors into a cohesive network growth model is challenging and also a topic for future investigation.

CONFLICT OF INTEREST

All authors declare no conflicts of interest related to this work.

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DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

REFERENCES

1. Chesnoy J. *Undersea Fiber Communication Systems*. London, UK: Academic Press, 2015.
2. McCurdy A, Shelton A, Burns B *et al*. Submarine telecoms industry report. In: *Submarine Telecoms Forum, Inc.*, Tech. Rep. 2018.
3. Xie Y, Wang C, Huang J. Structure and evolution of the submarine cable network of Chinese mainland. *J Geogr Sci* 2022;**32**:932–56.
4. TeleGeography. *Submarine Cable Map*. Available at: <https://www.submarinecablemap.com> (18 September 2025, date last accessed).
5. Bischof ZS, Fontugne R, Bustamante FE. Untangling the world-wide mesh of undersea cables. In: *Proceedings of the 17th ACM Workshop on Hot Topics in Networks*, HotNets '18, p. 78–84, New York, NY: Association for Computing Machinery, 2018.
6. Beaufils J-M. How do submarine networks web the world? *Optical Fiber Technol* 2000;**6**:15–32.
7. Malecki EJ, Wei H. A wired world: the evolving geography of submarine cables and the shift to Asia. *Ann Assoc Am Geographers* 2009;**99**:360–82.
8. Bononi A, Serena P, Antona J-C. A state-variable approach to submarine links capacity optimization. *J Lightwave Technol* 2021;**39**:5753–65.
9. Wang T, Wang X, Wang Z *et al*. Optimal tree topology for a submarine cable network with constrained internodal latency. *J Lightwave Technol* 2021;**39**:2673–83.
10. Wang T, Wang Z, Moran B *et al*. Submarine cable network design for regional connectivity. *IEEE/ACM Trans Netw* 2022;**30**:2480–92.
11. Wang X, Wang Z, Wang T *et al*. Designing cost-effective and reliable submarine communications cable path: lessons from the Tonga Volcano disaster. *IEEE Commun Mag* 2023;**61**:179–85.
12. Wang X, Wang Z, Zukerman M. Application of adaptive parallel fast marching method in automatic submarine cable path planning. *IEEE Trans Automat Sci Eng* 2024;**22**:9498–514.
13. Wang Z, Wang Q, Moran B *et al*. Optimal submarine cable path planning and trunk-and-branch tree network topology design. *IEEE/ACM Trans Netw* 2020;**28**:1562–72.
14. Sarif A, Akbar TAR, Kushartadi T, Apriono C. Design and analysis of multi-core fiber with SDM (Space Division Multiplex) technology in Singapore to Surabaya via Southern Sea of Java. In: *2022 9th International Conference on Electrical Engineering, Computer Science and Informatics*, p. 414–419. Jakarta, Indonesia: Institute of Advanced Engineering and Science (IAES), 2022.
15. TeleGeography Submarine Cable Map.
16. Wang Q, Wang Z, Guo J *et al*. Path planning of submarine cables. In: *2019 21st International Conference on Transparent Optical Networks (ICTON)*, p. 1–4. Piscataway, NJ, USA: IEEE, 2019.
17. Carisimo E, Wang CJ, Weaver M *et al*. A hop away from everywhere: a view of the intercontinental Long-haul infrastructure. *Proc ACM Meas Anal Comput Syst* 2023;**7**:1–26.
18. Livadariu I, Elmokashfi A, Smaragdakis G. Tracking submarine cables in the wild. *Comput Netw* 2024;**242**:110234.
19. Ramanathan A, Abdu Jyothi S. Nautilus: a framework for cross-layer cartography of submarine cables and IP links. *Proc ACM Meas Anal Comput Syst* 2023;**7**:1–34.
20. Ionescu M, Renaudier J, Ghazisaeidi A *et al*. Optimization of power efficient spatial division multiplexed submarine cables using adaptive transponders and machine learning. *J Lightwave Technol* 2022;**40**:1597–604.
21. Barabási A-L, Albert R. Emergence of scaling in random networks. *Science* 1999;**286**:509–12.
22. Chung F, Lu L, Dewey TG *et al*. Duplication models for biological networks. *J Comput Biol* 2003;**10**:677–87.